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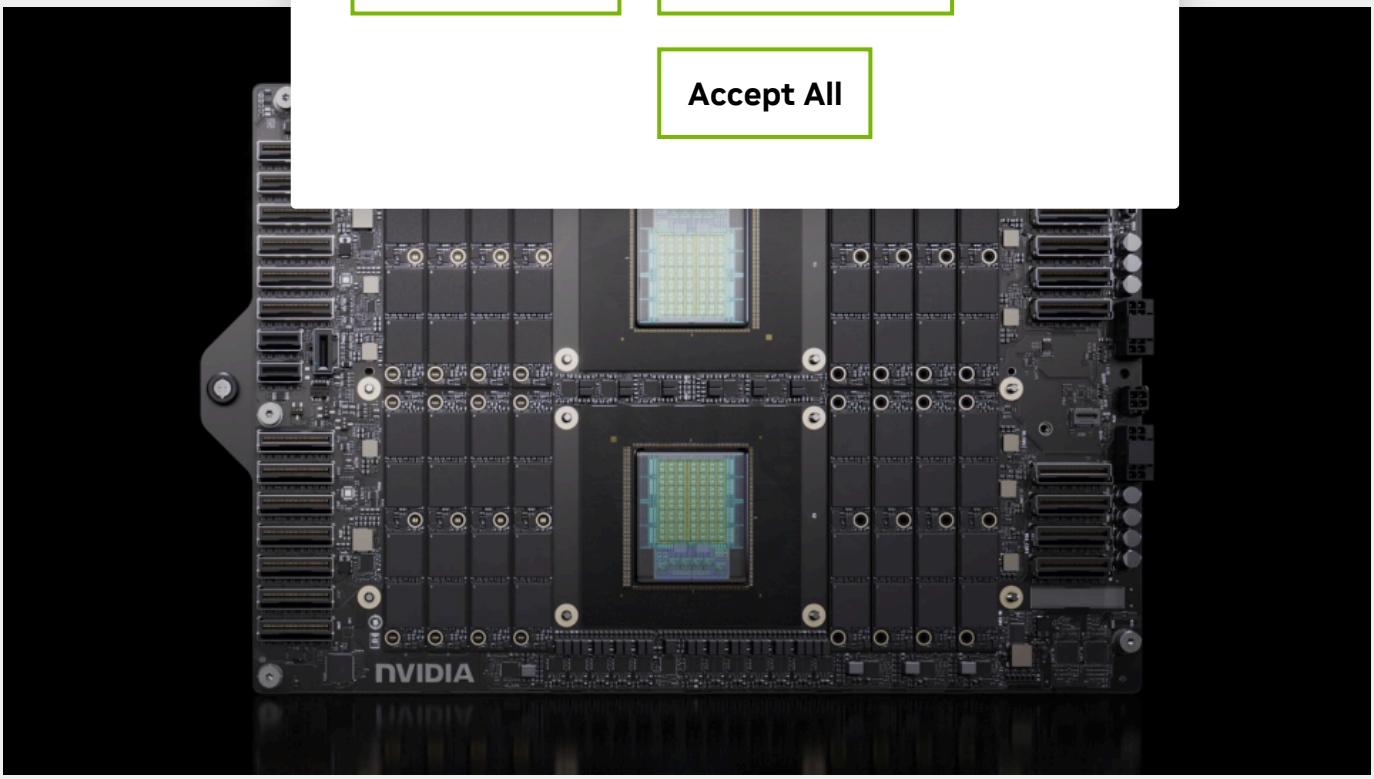
NVIDIA High Performance Bandwidth and Efficiency for AI Factories



Mar 16, 2026

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By Praveen Menon and Ivan Goldwasser

AI is evolving, and reasoning models are increasing token demand, placing new requirements on ev scale efficiently to maxim eators and users.

Modern GPUs opera ation, but system performanc an agentic loop–a classic exam v.

This dynamic is esp rning (RL) for training models wit gentic actions, which enab interpreters, and ot dboxes.

Both workloads cor al environments requi code quickly, similar to a worksta ls of these environments concurrently, creating large-scale throughput demands typical of server infrastructure.

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The NVIDIA Vera CPU is designed for modern AI workloads, with key design features including:

- **Extreme single-core performance:** Fast execution of individual tasks is critical, and performance must sustain under constant load with many concurrent users and agentic tasks.
- **High memory and fabric bandwidth per core:** To ensure consistent SLA under load that moves volumes of data efficiently for real-time analysis and context switching tasks.
- **Efficient rack-scale co-design:** AI factories must rapidly deploy and manage capacity to fulfill agentic demand while maximizing power efficiency.

Data centers built with Vera maximize AI infrastructure investments, whether Vera CPUs are directly connected to accelerators or performing tasks on standalone CPU capacity at the end of a wire.

Reinforcement learning requires models to constantly evaluate their outputs, recognizing which results succe lopment generate large amo then shipped to clusters d loop (see Figure 1).

These tasks span c data conversion, and oth current sandbox-like enviro e CPU core executes each light rated requests.

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Figure 1. A CPU Sandbox accelerating RL-based post-training and agentic calling

To maximize accelerator utilization and enforce rapid model iteration, the token generation and training phases of the cycle operate on a tight schedule (or policy). Often, some evaluation jobs running on a CPU finish jobs too late to influence the next step in the cycle. When this happens, it takes the model longer to learn to the same quality, and valuable tokens are wasted.

Agentic loops demand a unique blend of high single-core performance, massive data bandwidth, and deterministic execution with minimal tail latencies from the CPUs they employ.

TB/s of memory bandwidth, and 88 Olympus cores with NVIDIA Spatial Multithreading (SMT) for the task concurrency necessary for AI Factories.

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Figure 2. Vera CPU architecture

NVIDIA Olympus core

The need for higher-performance cores that support AI led to the NVIDIA Olympus core, the first fully custom data center CPU core from NVIDIA. Olympus debuts in Vera alongside the second generation of the NVIDIA Scalable Coherency Fabric (SCF), originally developed for the NVIDIA Grace CPU.

Built for sustained high Instruction Per Cycle (IPC) operation on memory-intensive workloads with control-flow logic, Olympus uses a 10-wide instruction fetch and decode frontend, and a neural branch predictor capable of evaluating two taken branches per cycle. It is fully compatible with the Arm v9.2 instruction set and existing software for high performance on Arm-based containers, binaries, libraries, and operating systems.

tail latency under heavy load. Traditional SMT relies on time-shared resources and frequent context switching between threads, introducing performance variation.

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NVIDIA Sca ory subsystem

The Vera CPU is bui nt dielets implementing mem he compute topology.

From the point of v ce to resources like other with uniform, high-throu local, avoiding unnecessa

The runtime paths rchestration, and control planes a implementations, the topology of the processor and the usage patterns of neighboring tasks being run on it must be considered ahead of time to maximize application performance. The design enables optimal performance without this style of tuning.

The second-generation SCF connects all 88 Olympus cores to a shared L3 cache and memory subsystem, delivering consistent latency and 3.4 TB/s of bisection bandwidth, enabling the Vera CPU to sustain over 90% of peak memory bandwidth under load. Each core is provisioned with up to 14 GB/s of memory bandwidth, roughly 3x the per-core rate of traditional data center CPUs—ensuring Extract-Transform-Load (ETL), real-time analytics, and memory-bound workloads maintain throughput when every core is active.

Feeding SCF is Vera's second-generation LPDDR5X memory subsystem, delivering up to 1.2 TB/s of total bandwidth at less than half the memory power of traditional DDR configurations and up to 1.5 TB of capacity—a 3x increase over the prior generation. Small Outline Compression-Attached Memory Modules (SOCAMM) brings low-power memory into the data center for the first time, replacing soldered memory with detachable, upgradable modules that combine LPDDR efficiency with server-class serviceability.

All these architectu gentic sandbox performan forms across compilers, scripting (Figure 3).

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Figure 3. NVIDIA Vera offers up to 1.5x higher sandbox performance compared to the competition

This advantage compounds across three dimensions. In RL post-training, a 1.5x faster sandbox returns evaluation results within tighter time windows, enabling models to capture the best gradient tokens and accelerating training cycles.

In agentic inference, it reduces users' wait time, improving accelerator utilization and easing pressure on KV cache offloading.

can learn from.

Agentic

Every AI Factory rec and tool
use. To unlock the p many AI
factory operators, t nters
designed for high-r

The new NVIDIA Ve n the same
planning constraint ucts being
deployed today.

With a capacity of r the capacity
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Each Vera CPU is connected with NVIDIA BlueField-4 SmartNICs containing dedicated Grace-based management cores, offloading networking tasks like security and management, and ensuring the most performant capacity in the system is fully available to agentic tasks.

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Figure 4. Rack-level efficiency across RL sandbox evaluation, ETL, and analytics under full system load

Vera platforms and configurations

In addition to the Vera CPU rack, NVIDIA has engineered a complete family of Vera-based platforms for the diverse workloads of modern AI factories. By delivering many choices of densities, cooling capabilities, configurations, and form factors, Vera's design and system partners are enabling rapid deployment and capacity build-out, adaptable to the constraints of space available in any data center facility.

Platform	Description	Scenarios

	NVIDIA NVLink scale-up fabric.	throughput inference.
NVIDIA Vera CPU Rack		ucture, rs, data nd CPU-
Single and dual-socket Vera platforms		re, cs, storage, GPU- and AI
NVIDIA HGX Rubin NVL8		ical cs, and ployments.

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Figure 5. Vera CPU at AI factory scale

Platform availability

Vera systems will be ... vo, and Supermicro, in the s ... tails.

Learn More about t ...

NVIDIA Vera performanc ... iety of workloads, including cod ... nd graph.

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Praveen is a senior technical marketing engineer focused on accelerated computing platforms in the data center at NVIDIA. Previously, Praveen held various roles across marketing, performance, and silicon engineering. He holds a master's degree in electrical and computer engineering from the University of Arizona

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Ivan leads product marketing for the Data Center CPU products for NVIDIA. Previously, Ivan worked in various marketing and strategy roles in the technology sector. Ivan has an MBA from Georgetown's McDonough School of Business and a bachelor's degree in chemical engineering from Texas A&M University.

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